

Properties 17.1

- i. If $AB=BA$, then $e^{At} e^{Bt} = e^{Bt} e^{At} = e^{(A+B)t}$
- ii. e^{At} is invertible with inverse $(e^{At})^{-1} = e^{-At}$
- iii. $\frac{d}{dt} e^{At} = A e^{At}$
- iv. $\int e^{At} dt = A^{-1} e^{At} + C$ (provided by A^{-1} exists)

Proofs:

$$\begin{aligned}
 \text{i. } (e^{At})(e^{Bt}) &= \left(\sum_i \frac{A^i t^i}{i!} \right) \left(\sum_j \frac{B^j t^j}{j!} \right) \\
 &= \sum_i \left(\sum_{j=0}^i \frac{A^j B^{i-j}}{j! (i-j)!} \right) \frac{t^i}{i!} \\
 &= \sum_i \left(\sum_{j=0}^i \binom{i}{j} A^j B^{i-j} \right) \frac{t^i}{i!}
 \end{aligned}$$

If $AB=BA$ then by the Binomial Theorem

$$(A+B)^i = \sum_{j=0}^i \binom{i}{j} A^j B^{i-j}$$

$$\begin{aligned}
 (A+B)^2 &= (A+B)(A+B) = A^2 + AB + BA + B^2 \\
 &= A^2 + 2AB + B^2 \quad (AB=BA)
 \end{aligned}$$

ii. Follows from 17.1.i. as A and $-A$ commute

$$\begin{aligned}\text{iii. } \frac{d}{dt} e^{At} &= \frac{d}{dt} \sum_{n=0}^{\infty} \frac{A^n t^n}{n!} = \sum_{n=1}^{\infty} \frac{A^n t^{n-1}}{(n-1)!} \\ &= A \sum_{n=1}^{\infty} \frac{A^{n-1} t^{n-1}}{(n-1)!} \\ &= A e^{At}\end{aligned}$$

$$\text{iv. } \frac{d}{dt} A^{-1} e^{At} = A^{-1} \frac{d}{dt} e^{At} = A^{-1} A e^{At} = e^{At}$$

Solving $\vec{x}' = A\vec{x}$

Rewriting as $\vec{x}' - A\vec{x} = \vec{0}$ and multiplying both sides by e^{-At} we get

$$\begin{aligned}\vec{0} &= e^{-At} \cdot \vec{x}' - A e^{-At} \cdot \vec{x} \\ &= \frac{d}{dt} e^{-At} \cdot \vec{x}\end{aligned}$$

Now integrating both sides we get

$$e^{-At} \cdot \vec{x} = \vec{c}$$

or $\vec{x} = e^{At} \cdot \vec{c}$.

Theorem 17.2 If $A = S \Lambda S^{-1}$, then

$$\vec{x}' = A \vec{x}, \quad \vec{x}(0) = \vec{x}_0$$

has solution $\vec{x} = S e^{\Lambda t} S^{-1} \cdot \vec{x}_0$ $e^{A \cdot 0} = I$

Proof: letting $\vec{x} = e^{At} \cdot \vec{c}$ and setting $A = S \Lambda S^{-1}$ we have that $\vec{x} = S e^{\Lambda t} S^{-1} \cdot \vec{c}$ by Theorem 16.7. Now setting $\vec{x}(0) = \vec{x}_0$ we have that

$$\vec{x}_0 = S I S^{-1} \vec{c}$$

and thus $\vec{c} = \vec{x}_0$

Exercise :

1) Solve the linear system

$$x_1' = 2x_1 - x_2, \quad x_1(0) = -3$$

$$x_2' = 6x_1 + 9x_2, \quad x_2(0) = 4$$

Setting $A = \begin{bmatrix} 2 & -1 \\ 6 & 9 \end{bmatrix}$ we find e^{At}

by diagonalizing A :

e-values: $\lambda = 3, 8$

e-vectors: $\lambda = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -6 \end{bmatrix}$

Thus $e^{At} = S e^{\Lambda t} S^{-1}$ where

$$S = \begin{bmatrix} 1 & 1 \\ -1 & -6 \end{bmatrix} \quad \text{and} \quad e^{\Lambda t} = \begin{bmatrix} e^{3t} & \\ & e^{8t} \end{bmatrix}$$

Now by Theorem 17.2

$$\vec{x} = S e^{\Lambda t} S^{-1} \vec{x}_0$$

$$= \begin{bmatrix} 1 & 1 \\ -1 & -6 \end{bmatrix} \begin{bmatrix} e^{3t} & \\ & e^{8t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -6 \end{bmatrix}^{-1} \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ -1 & -6 \end{bmatrix}^{-1} = \frac{1}{5} \begin{bmatrix} -6 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 1 & 1 \\ -1 & -6 \end{bmatrix} \begin{bmatrix} e^{3t} & \\ & e^{8t} \end{bmatrix} \begin{bmatrix} 6 & 1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 1 & 1 \\ -1 & -6 \end{bmatrix} \begin{bmatrix} e^{3t} & \\ & e^{8t} \end{bmatrix} \begin{bmatrix} -14 \\ -1 \end{bmatrix}$$

Solving $\vec{x}' - A\vec{x} = \vec{u}(t)$:

Once again multiplying both sides by e^{-At}
we obtain

$$\frac{d}{dt} e^{-At} \cdot \vec{x} = e^{-At} \cdot \vec{u}(t)$$

and integrating both sides we have that

$$e^{-At} \cdot \vec{x} = \int e^{-At} \cdot \vec{u}(t) dt + \vec{c}$$

and thus the general solution is

$$\vec{x} = e^{At} \left[\int \underbrace{e^{-At} \cdot \vec{u}(t)}_{\text{IBP!}} dt + \vec{c} \right]$$

Exercise :

2) Find a general solution

$$\vec{x}' - \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix} \vec{x} = \begin{bmatrix} t \\ t+3 \end{bmatrix}$$

Computing $\int e^{-At} \begin{bmatrix} t \\ t+3 \end{bmatrix} dt$ using integration
by parts

u	dv
$(t, t+3)$	e^{-At}
$(1, 1)$	$-(A^{-1})e^{-At}$
$(0, 0)$	$(A^{-1})^2 e^{-At}$

we get
$$e^{-At} \left[-A^{-1} \begin{bmatrix} t \\ t+3 \end{bmatrix} - A^{-2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right]$$

and thus
$$\vec{x} = e^{At} \left[\int e^{-At} \begin{bmatrix} t \\ t+3 \end{bmatrix} dt + \vec{c} \right]$$

$$= -A^{-1} \begin{bmatrix} t \\ t+3 \end{bmatrix} - A^{-2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{At} \vec{c}$$

Now as $A^{-1} = \frac{1}{7} \begin{bmatrix} -5 & 4 \\ 3 & -1 \end{bmatrix}$ and

$$A^{-2} = \frac{1}{7} \begin{bmatrix} -5 & 4 \\ 3 & -1 \end{bmatrix} = \frac{1}{49} \begin{bmatrix} 37 & -24 \\ -18 & 13 \end{bmatrix}$$

we have that

$$\vec{x} = -\frac{1}{7} \begin{bmatrix} -5 & 4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} t \\ t+3 \end{bmatrix} - \frac{1}{49} \begin{bmatrix} 37 & -24 \\ -18 & 13 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{At} \vec{c}$$

$$= \frac{1}{49} \begin{bmatrix} 7t - 97 \\ -14t + 26 \end{bmatrix} + e^{At} \vec{c}$$

Finally as A has e-values of $\lambda = -1, 7$
and e-vectors of $\vec{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$$\vec{x} = \begin{bmatrix} \frac{t}{7} - \frac{97}{49} + 2c_1 e^{-t} + 2c_2 e^{7t} \\ -\frac{2t}{7} + \frac{26}{49} - c_1 e^{-t} + 3c_2 e^{7t} \end{bmatrix}$$

end of exam 2 material

Defective E-values and Generalized E-vectors

Recall that given e-value λ_i of A , the algebraic multiplicity of λ_i is the order of zero of $p_A(\lambda)$ at λ_i and the geometric multiplicity is the dimension of $\text{null}(A - \lambda_i I)$

Def 17.3 An e-value is defective (degenerate) iff the geometric mult. is strictly less than the algebraic mult.

For example: If $p_A(\lambda) = (\lambda - \lambda_i)^{n_i} q(\lambda)$, then λ_i is defective iff $\text{nullity}(A - \lambda_i I) < n_i$.

Exercise:

3) Show that the matrix

$$A = \begin{bmatrix} 3 & 2 \\ 0 & 3 \end{bmatrix}$$

has a defective e-value

Since $|A - \lambda I| = (3 - \lambda)^2$, $\lambda = 3$ has alg. mult. of 2. However as $A - 3I = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$ is rank one, nullity $(A - 3I) = 1 < 2$.

Theorem 17.4 An e-value is defective iff the order of the corresponding zero of the minimal polynomial is not simple. (not order one)

Proof: Let $A \in \mathbb{C}^{n \times n}$ have charact polynomial of $p = (\lambda - \lambda_0)^n$ and minimal polynomial of $\mu = (x - \lambda_0)^2$. Now let $\{\vec{v}_1, \dots, \vec{v}_k\}$ be a basis for $\text{null}(A - \lambda_0 I)$. Since $\text{null}((A - \lambda_0 I)^2) = \mathbb{C}^n$, extend $\{\vec{v}_1, \dots, \vec{v}_k\}$ to a basis $\mathcal{B}$ of $\mathbb{C}^n (= \text{null}(A - \lambda_0)^2)$.

By def. of the minimal polynomial

$A - \lambda_0 I$ is not the zero matrix and thus pick $\vec{w} \in \mathcal{B}$ such that $(A - \lambda_0 I)\vec{w} \neq \vec{0}$.

Hence $|\mathcal{B}| > k$ and thus $k < n$.

Corollary 17.5 A matrix A is diagonalizable
iff μ_A has only simple zeros.

Theorem 17.6 let λ_i be e-value of A
and let $\vec{v}_{i_1}, \dots, \vec{v}_{i_k}$ be non-zero vectors s.t.
 $A \vec{v}_{i_1} = \lambda_i \vec{v}_{i_1}$ ($\vec{v}_{i_1}$ is an e-vector) and
$$(A - \lambda_i I) \vec{v}_{i_j} = \vec{v}_{i_{j-1}}$$

Then $\vec{v}_{i_1}, \dots, \vec{v}_{i_k}$ are linearly independent
 $\vec{v}_{i_j} \in \text{null } (A - \lambda_i I)^j$ for all $1 \leq j \leq k$.

let $\vec{v}_{i_1}, \dots, \vec{v}_{i_k}$ be as above and assume
that $\vec{v}_{i_k} = \sum_{j=1}^{k-1} c_j \vec{v}_{i_j}$ where $\vec{v}_{i_1}, \dots, \vec{v}_{i_{k-1}}$
are lin. ind. and c_j are not all zero. Now

$$\vec{v}_{i_{k-1}} = (A - \lambda_i I) \vec{v}_{i_k} = \sum_{j=1}^{k-1} c_j \underbrace{(A - \lambda_i I) \vec{v}_{i_j}}_{\vec{v}_{i_{j-1}}}$$

$$= \sum_{j=2}^{k-1} c_j \vec{v}_{i_{j-1}}$$

or that $\vec{v}_{i_{k-1}} \in \text{span}(\vec{v}_{i_1}, \dots, \vec{v}_{i_{k-2}})$

which contradicts the assumption that

$\vec{v}_{i_1}, \dots, \vec{v}_{i_{k-1}}$ was lin. ind.

Def 17.7 let λ be an e-value of A .

Then $\vec{v}$ is a generalized e-vector of rank

m iff $(A - \lambda I)^m \vec{v} = \vec{0}$ but

$(A - \lambda I)^j \vec{v} \neq \vec{0}$ for $j < m$.

Jordan Chain

let λ_i be defective and let $\vec{v}_{i_1}$ be s.t.

$A \vec{v}_{i_1} = \lambda_i \vec{v}_{i_1}$ ($\vec{v}_{i_1}$ is an e-vector). Then

the generalized e-vectors of A corresponding to λ_i and $\vec{v}_{i_1}$ are given by

Start with e-vector $\vec{v}_{i_1}$

Set $(A - \lambda_i I) \vec{v}_{i_2} = \vec{v}_{i_1}$

Set $(A - \lambda_i I) \vec{v}_{i_3} = \vec{v}_{i_2}$

$\vdots$

Set $(A - \lambda_i I) \vec{v}_{i_k} = \vec{v}_{i_{k-1}}$

$\vdots$

end when $(A - \lambda_i I) \vec{v}_{i_m} = \vec{v}_{i_{m-1}}$

is not solvable (no solutions)

Exercise:

4) Find all e-vectors and generalized
of the matrix $A = \begin{bmatrix} 1 & -1 \\ 1 & 3 \end{bmatrix}$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & -1 \\ 1 & 3-\lambda \end{vmatrix} = \lambda^2 - 4\lambda + 4 \\ = (\lambda - 2)^2 \\ \lambda = 2 \text{ (x2)}$$

$$\mu = (\lambda - 2)^2 \text{ as well}$$

$$A - 2I = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \vec{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Setting $(A - 2I)\vec{v}_{1,2} = \vec{v}_{1,1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\left[\begin{array}{cc|c} -1 & -1 & 1 \\ 1 & 1 & -1 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} -1 & -1 & 1 \\ 0 & 0 & 0 \end{array} \right]$$

$$a + b = -1$$

Set $b = 0$ so that

$$a = -1$$

we pick $\vec{v}_{1,2} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$ and thus

The generalized e-vectors corresponding

to $\lambda = 2$ are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \end{bmatrix}$


not unique of course